

Subject: Burgers Equation with Viscous and Artificial-Viscosity Shock

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Burgers Equation with Viscous and Artificial-Viscosity Shock

That changes the interpretation substantially. With a delta formulation + Newton iteration, we can write down the discrete nonlinear problem much more precisely.

Suppose at time level $n+1$ you solve

$$\mathbf{R}(\mathbf{u}^{n+1}) = 0 \quad (1)$$

by Newton iteration. At Newton iteration k ,

$$\mathbf{J}(\mathbf{u}^{(k)})\delta\mathbf{u} = -\mathbf{R}(\mathbf{u}^{(k)}), \quad (2)$$

followed by

$$\mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} + \delta\mathbf{u}. \quad (3)$$

Your matrix $\mathbf{A}$ is therefore the Newton Jacobian, not necessarily the physical diffusion operator itself. This is the important distinction that was missing in the previous derivation.

Your artificial-viscosity Jacobian

You have

$$A_{i,i-1} = \frac{1}{8} - c_\Psi, \quad (4)$$

$$A_{i,i} = \frac{6}{8} + 2c_\Psi, \quad (5)$$

$$A_{i,i+1} = \frac{1}{8} - c_\Psi. \quad (6)$$

The artificial part of the Jacobian is

$$c_\Psi[-1, 2, -1]. \quad (7)$$

For the **delta variable**,

$$\delta u_{i-1} - 2\delta u_i + \delta u_{i+1} = \Delta x^2 \frac{\partial \delta u}{\partial x} + \frac{1}{12} \Delta x^4 \frac{\partial^4 \delta u}{\partial x^4}. \quad (8)$$

Therefore the artificial-viscosity contribution to the Newton system is

$$\boxed{-c_\Psi \Delta x^2 \frac{\partial \delta u}{\partial x} - \frac{c_\Psi}{12} \Delta x^4 \frac{\partial^4 \delta u}{\partial x^4} + \dots} \quad (9)$$

and this is a contribution to the **Jacobian**, not directly $\nu_{art} u_{xx}$.

The crucial point about your c_E

You have

$$c_E = \frac{\pi}{2} c_\Psi^2 \quad (10)$$

and

$$rhs_i = \frac{\pi}{4} c_\Psi^2 \Delta x^2 |u_{xx}| \quad (11)$$

Because this appears in the **Newton residual/RHS**, it represents a nonlinear residual contribution. If the residual contains

$$\mathbf{R}_\Psi(\mathbf{u}) = \frac{\pi}{4} c_\Psi^2 \Delta x^2 |u_{xx}| \quad (12)$$

then its Newton linearisation is

$$\delta \mathbf{R}_\Psi = \frac{\pi}{4} c_\Psi^2 \Delta x^2 \text{sign}(u_{xx}) \delta u_{xx} \quad (13)$$

Thus the Jacobian contribution associated with this term should be proportional to

$$\boxed{\frac{\pi}{4} c_\Psi^2 \Delta x^2 \text{sign}(u_{xx}) \delta u_{xx}} \quad (14)$$

This is quite different from a conventional constant viscosity.

What your scheme is effectively doing

Ignoring higher-order terms, your Newton equation contains artificial terms of the form

$$-c_\Psi \Delta x^2 \delta u_{xx} + \frac{\pi}{4} c_\Psi^2 \Delta x^2 \text{sign}(u_{xx}) \delta u_{xx} \quad (15)$$

Therefore the effective coefficient associated with the second derivative is approximately

$$\boxed{\nu_{eff} \propto c_\Psi \Delta x^2 - \frac{\pi}{4} c_\Psi^2 \Delta x^2 \text{sign}(u_{xx})} \quad (16)$$

up to the sign convention of your residual.

That sign convention is important: depending on whether you define

$$R = F(u) - rhs \quad (17)$$

or

$$R = rhs - F(u) \quad (18)$$

the signs reverse.

So I would **not yet use this formula to choose c_Ψ** .

What I now need

Since you have told me that the method is a delta/Newton formulation, the missing piece is very small. Please give me the actual Newton equation, something like

$$\mathbf{A}\delta = rhs; \tag{19}$$

and preferably the definition of rhs immediately before this. Then I can derive exactly:

$$\boxed{R(u)} \quad \boxed{J(u) = \frac{\partial R(u)}{\partial u}} \tag{20}$$

and consequently the continuous modified equation

$$u_u + uu_x = \nu_{eff}(u, u_x, u_{xx})u_{xx} + HOT \tag{21}$$

That will also let us determine whether your current c_E term is **actually adding viscosity or inadvertently reducing it**, which is particularly relevant to the wiggles you observed for $a = 1, b = -1$.